

can show behavior analogous to thermodynamic transitions.

In nested-sampling language, abrupt changes in the structure of $L(X)$ can signal transitions between large-volume/low-likelihood and small-volume/high-likelihood regions.

So multimodality, metastability, and phase coexistence have quite literal Bayesian analogues.

The interpretation I find most useful

For the figures we have just constructed, I would explain the analogy this way:

Treat $-\log L$ as an energy. Each likelihood contour is then an energy contour. The prior volume X is the cumulative phase-space volume enclosed by that contour. Nested sampling systematically compresses this accessible phase-space volume, thereby reconstructing the density-of-states information needed to calculate the partition function—i.e. the Bayesian evidence.

And the particularly elegant chain is

$$L(\theta) \longleftrightarrow E(\theta) = -\log L(\theta) \longleftrightarrow X(E) \longleftrightarrow g(E) \longleftrightarrow Z.$$

That is much closer than a loose analogy: nested sampling can genuinely be viewed as a statistical-mechanics **density-of-states/free-energy algorithm**.



Please give more detail on the derivation of the beta distribution for t.

Certainly. The Beta distribution is not an extra assumption in nested sampling; it follows directly from an **order-statistics argument** for the N_{live} live points.

Suppose that at iteration $i - 1$, nested sampling has reached the likelihood threshold L_{i-1} , corresponding to remaining prior volume

$$X_{i-1}.$$

The live points are assumed to be independent samples from the prior subject to

$$L(\theta) > L_{i-1}.$$

Define the shrinkage factor

$$t_i \equiv \frac{X_i}{X_{i-1}},$$

where X_i is the prior volume after we discard the lowest-likelihood live point.

The claim is

$$t_i \sim \text{Beta}(N_{\text{live}}, 1).$$

Here is the derivation.

1. Why is a live point uniform in X ?

Recall the definition

$$X(\lambda) = \int_{L(\theta) > \lambda} \pi(\theta) d\theta.$$

So X is literally a cumulative prior-probability coordinate. As the likelihood threshold rises,

$$1 \geq X \geq 0.$$

If we draw

$$\theta \sim \pi(\theta),$$

then the corresponding prior-volume coordinate $X(\theta)$ is uniformly distributed between 0 and 1. This is essentially the probability-integral transform.

Now suppose we have already imposed

$$L > L_{i-1}.$$

That confines us to

$$0 < X < X_{i-1}.$$

A point drawn from the constrained prior therefore has

$$X_j \mid X_{i-1} \sim \text{Uniform}(0, X_{i-1}).$$

It is convenient to divide out the current volume:

$$u_j = \frac{X_j}{X_{i-1}}.$$

Then

$$u_j \sim \text{Uniform}(0, 1).$$

This is the crucial simplification.

2. Which live point gets discarded?

Suppose there are

$$N = N_{\text{live}}$$

live points. In X -coordinates they are

$$u_1, u_2, \dots, u_N$$

with each

$$u_j \sim U(0, 1).$$

Remember that small X means high likelihood. Schematically,

$$X \downarrow \iff L \uparrow.$$

Therefore the **lowest-likelihood** live point has the **largest** X .

So if that point establishes the next likelihood threshold,

$$t_i = \frac{X_i}{X_{i-1}}$$

is just

$$t_i = \max(u_1, \dots, u_N).$$

Thus the nested-sampling shrinkage problem has become a very elementary order-statistics problem:

What is the distribution of the largest of N independent uniform random variables?

3. CDF of the maximum

Let

$$T = \max(u_1, \dots, u_N).$$

We first calculate its cumulative distribution,

$$P(T < t).$$

For the largest value to be below t , **every one of the N points must be below t** :

$$T < t \iff u_1 < t, \dots, u_N < t.$$

Since the variables are independent,

$$P(T < t) = \prod_{j=1}^N P(u_j < t).$$

For a uniform random variable,

$$P(u_j < t) = t, \quad 0 < t < 1.$$

Therefore

$$P(T < t) = t^N.$$

That is the CDF.

Differentiate to obtain the probability density:

$$p_T(t) = \frac{d}{dt}t^N = Nt^{N-1}, \quad 0 < t < 1.$$

Now compare with the general Beta density,

$$p(t) = \frac{1}{B(a,b)} t^{a-1} (1-t)^{b-1}.$$

For

$$a = N, \quad b = 1,$$

we have

$$B(N, 1) = \frac{1}{N},$$

giving

$$p(t) = Nt^{N-1}.$$

Hence

$$t_i \sim \text{Beta}(N_{\text{live}}, 1).$$

4. Why is t so close to one?

For, say,

$$N_{\text{live}} = 100,$$

the density is

$$p(t) = 100t^{99}.$$

This is enormously concentrated near $t = 1$.

That makes intuitive sense. If you scatter 100 points uniformly between 0 and 1, the largest one will normally be very close to 1.

The mean is

$$E[t] = \frac{N}{N+1}.$$

Thus for $N = 100$,

$$E[t] = \frac{100}{101} \simeq 0.9901.$$

At first that sounds like nested sampling is making extremely little progress per iteration. But the important quantity is actually $\log X$, not X itself.

5. Why does $X_i \simeq e^{-i/N}$?

Start from

$$X_i = t_i X_{i-1}.$$

After i iterations,

$$X_i = t_i t_{i-1} \cdots t_1 X_0.$$

Since $X_0 = 1$,

$$X_i = \prod_{j=1}^i t_j.$$

Taking logarithms,

$$\log X_i = \sum_{j=1}^i \log t_j.$$

Now calculate the expectation of $\log t$:

$$E[\log t] = \int_0^1 N t^{N-1} \log t \, dt.$$

Using

$$\int_0^1 t^{N-1} \log t \, dt = -\frac{1}{N^2},$$

we get

$$E[\log t] = -\frac{1}{N}.$$

Therefore

$$E[\log X_i] = \sum_{j=1}^i E[\log t_j] = -\frac{i}{N},$$

so the characteristic trajectory is

$$X_i \sim e^{-i/N_{\text{live}}}.$$

This explains why nested sampling is naturally thought of as moving in **log prior volume**.

6. There is an even nicer form of the distribution

Define

$$s = -\log t.$$

Then

$$t = e^{-s}.$$

From

$$p_t(t) = Nt^{N-1}$$

and

$$\left| \frac{dt}{ds} \right| = e^{-s},$$

we obtain

$$p_s(s) = N(e^{-s})^{N-1}e^{-s} = Ne^{-Ns},$$

for $s \geq 0$.

So

$$-\log t \sim \text{Exponential}(N).$$

This immediately gives

$$E[-\log t] = \frac{1}{N},$$

and

$$\text{Var}(\log t) = \frac{1}{N^2}.$$

Consequently, after i iterations,

$$\log X_i = -\sum_{j=1}^i s_j,$$

so

$$E[\log X_i] = -\frac{i}{N},$$

and

$$\text{Var}(\log X_i) = \frac{i}{N^2}.$$

Thus the typical spread is

$$\sigma_{\log X_i} = \frac{\sqrt{i}}{N}.$$

This is the origin of the common shorthand

$$\log X_i \simeq -\frac{i}{N} \pm \frac{\sqrt{i}}{N}.$$

7. Why can we repeat this argument at every iteration?

There is a subtle but important point here.

Initially the N points are uniform on

$$0 < X < X_{i-1}.$$

Suppose their maximum occurs at X_i . Once we condition on that maximum, the surviving $N - 1$ points are uniformly distributed inside

$$0 < X < X_i.$$

Nested sampling then replaces the discarded point by drawing a new point from precisely that same constrained region,

$$L > L_i \iff X < X_i.$$

In the **idealized nested-sampling algorithm**, where that replacement is an independent exact draw from the constrained prior, we once again have N independent uniform points within the new volume.

Therefore the same order-statistics argument starts over:

$$\frac{X_{i+1}}{X_i} \sim \text{Beta}(N, 1).$$

So the shrinkage factors can be treated as iid Beta random variables in ideal nested sampling.

In a practical implementation, constrained sampling is usually accomplished with MCMC, slice sampling, ellipsoidal sampling, etc., so imperfect independence can modify this ideal picture.

8. A small numerical example

Take only

$$N_{\text{live}} = 4.$$

Suppose, unbeknownst to the algorithm, the four live points happen to have normalized prior-volume coordinates

$$0.12, \quad 0.37, \quad 0.61, \quad 0.83.$$

Since large X means low likelihood, nested sampling eliminates the point at

$$t = 0.83.$$

Thus

$$X_i = 0.83X_{i-1}.$$

On another realization we might obtain

$$0.20, \quad 0.43, \quad 0.88, \quad 0.95,$$

and then

$$t = 0.95.$$

The actual shrinkages vary from realization to realization. The Beta distribution

$$p(t) = 4t^3$$

describes precisely those fluctuations.

This connects directly to the issue we discussed earlier about the diagrams: the **actual X_i 's in a run are irregular**, while the statistical law governing their irregularity is

$$t_i = \frac{X_i}{X_{i-1}} \sim \text{Beta}(N_{\text{live}}, 1).$$

The deterministic-looking approximation

$$X_i \simeq e^{-i/N_{\text{live}}}$$

is only the typical log-volume trajectory through that stochastic process.

One particularly compact way to remember the whole derivation is

$$\begin{array}{l} N \text{ live points} \\ \Downarrow \\ u_1, \dots, u_N \stackrel{\text{iid}}{\sim} U(0, 1) \\ \Downarrow \\ t = \max(u_1, \dots, u_N) \\ \Downarrow \\ P(t < T) = T^N \\ \Downarrow \\ p(t) = Nt^{N-1} \\ \Downarrow \\ t \sim \text{Beta}(N, 1). \end{array}$$

And the perhaps even more useful equivalent statement is

$$-\log t \sim \text{Exp}(N_{\text{live}}).$$

That form makes the approximately constant rate of progress in $\log X$ especially transparent.



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